

Code Before and After the Recursive Call

The code in a recursive case can be split into two parts: the code before the recursive call and the code after the recursive call. (If there are two recursive calls in the recursive case, such as with the Fibonacci sequence example in Chapter 2, there will be a before, a between, and an after. But let's keep it simple for now.)

The important thing to know is that reaching the base case doesn't necessarily mean reaching the end of the recursive algorithm. It only means the base case won't continue to make recursive calls.

For example, consider this *countDownAndUp.py* program whose recursive function counts from any number down to zero, and then back up to the number:

Python

```
def countDownAndUp(number):  
    ❶ print(number)  
    if number == 0:  
        # BASE CASE  
        ❷ print('Reached the base case.')  
        return  
    else:  
        # RECURSIVE CASE  
        ❸ countDownAndUp(number - 1)  
        ❹ print(number, 'returning')  
        return
```

❺ countDownAndUp(3)

Here is the equivalent *countDownAndUp.html* program:

JavaScript

```
<script type="text/javascript">  
function countDownAndUp(number) {  
    ❶ document.write(number + "<br />");  
    if (number === 0) {  
        // BASE CASE
```